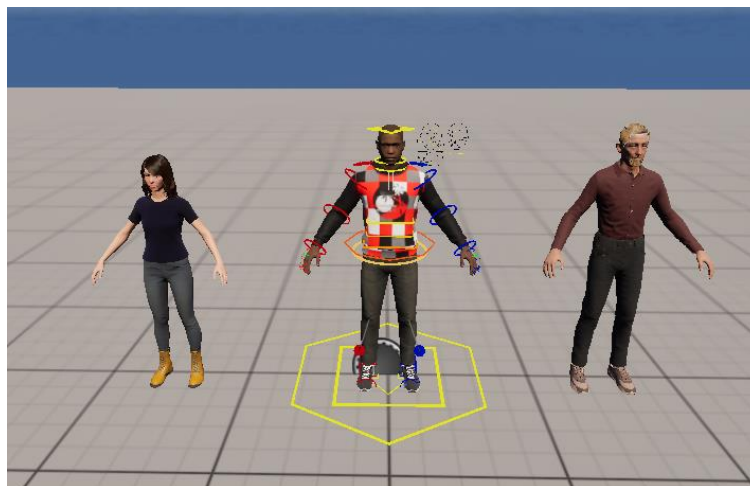
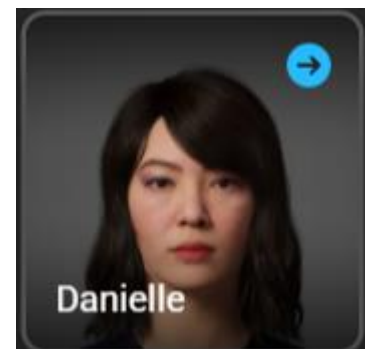
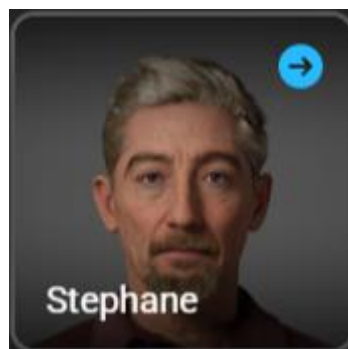
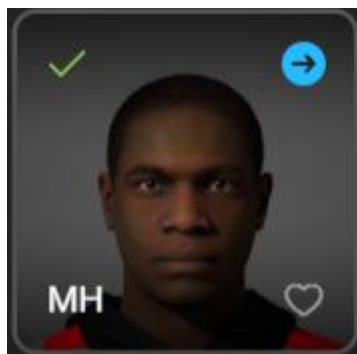
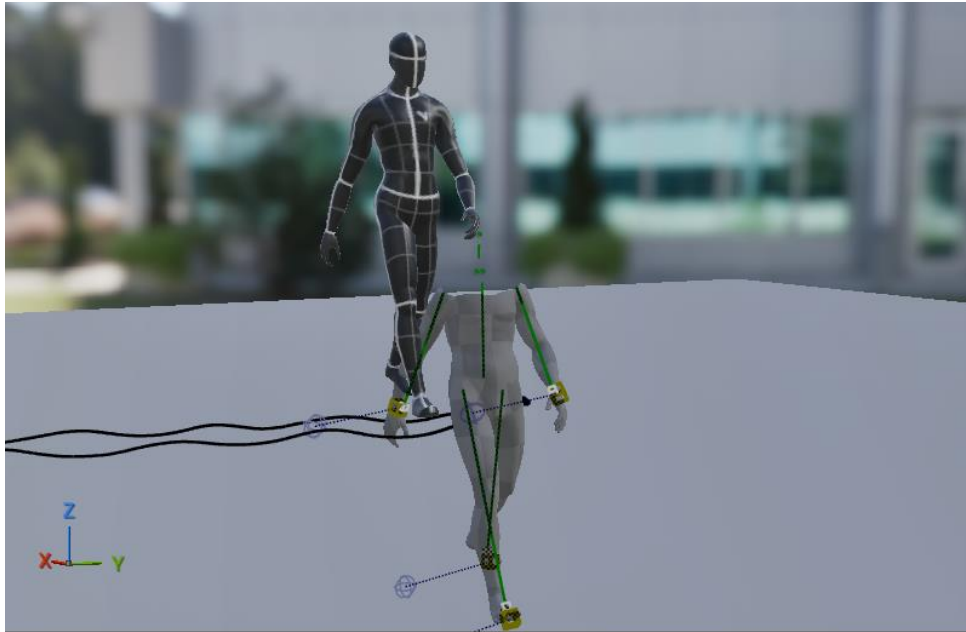

Assignment #04: Motion Editing**Demo 1 - Retargeting & Character Diversity**

This demo focuses on retargeting a single motion capture walking sequence to three distinct MetaHuman characters in Unreal Engine and then refining the motion to better suit each character's physical characteristics. The source animation was extracted from the provided Vicon motion capture dataset and imported into Unreal Engine, along with its skeleton. The animation was transferred from the source skeleton to three MetaHuman characters via Unreal Engine's IK-based retargeting system: a custom avatar created in a previous AR module (young, tall male), the Danielle MetaHuman (young, short female), and the Stephane MetaHuman (older, tall male).



Each MetaHuman required the creation of a compatible retargeting setup, including IK Rig and IK Retargeter assets. Specific attention was given to aligning the retarget poses between the source and target skeletons to ensure that joint orientations, especially in the shoulders and hips, were correctly matched prior to animation transfer. Once the retargeting pipeline was set up, each character's walking animation was exported and tested.



Following retargeting, motion edits were applied to each character individually to better reflect differences in body type and physical appearance. The custom self-avatar was largely unaltered after retargeting to serve as a baseline reference. The posture and motion pattern of the Stephane MetaHuman, who represents an older character, were subtly modified. A slight vertical bobbing motion was added to the stride, and the posture was altered to appear less rigid, implying decreased physical stability and a more natural ageing effect. In contrast, the Danielle MetaHuman needed to adjust her stride length. The stride was lengthened, and the timing was adjusted to better match the shorter character proportions, preventing the motion from appearing overly constrained.



Results

The retargeting process successfully transferred the original motion capture animation to all three MetaHumans while maintaining the overall structure and timing of the walk cycle. However, differences in body proportions resulted in noticeable variations in how the animation appeared between characters. The unedited retarget on the custom MetaHuman provided a consistent baseline, whereas the edited versions had better visual coherence when matched to each character's physical attributes.

After making posture and slight bobbing adjustments, the Stephane MetaHuman moved more like a character. The added vertical movement contributed to a sense of weight and age, setting it apart from the baseline animation. Similarly, stride and speed adjustments helped the Danielle MetaHuman reduce the visual mismatch between limb length and step distance, resulting in a more realistic walk cycle.



Evaluation and Reflection

While the retargeting process was effective in transferring the motion data, several limitations were observed. Most notably, foot sliding persisted across all characters, despite changes to stride length and animation timing. Although these modifications reduced the severity of the problem, they did not fully eliminate it, indicating limitations in the retargeting accuracy and the need for more advanced techniques such as foot locking or inverse kinematics constraints.

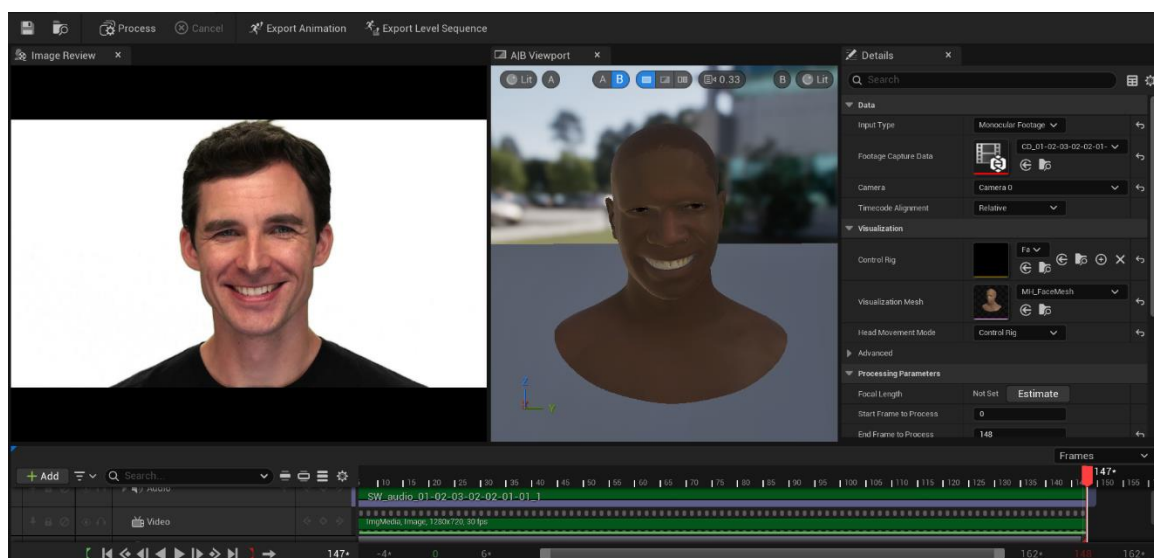
In addition, foot penetration was observed, with the feet occasionally clipping into the ground plane during the walk cycle. This issue is most likely caused by discrepancies between the original motion capture data, and the target character proportions, as well as a lack of precise ground contact correction during retargeting.

Despite these limitations, the demo shows how retargeted motion can be meaningfully adapted to different characters using relatively simple edits. The contrast between the three MetaHumans demonstrates how body type, age, and proportion influence perception of motion, as well as how targeted adjustments can improve realism. The findings also show that retargeting alone is insufficient for high-quality animation and must be combined with motion editing to achieve realistic character-specific movement.

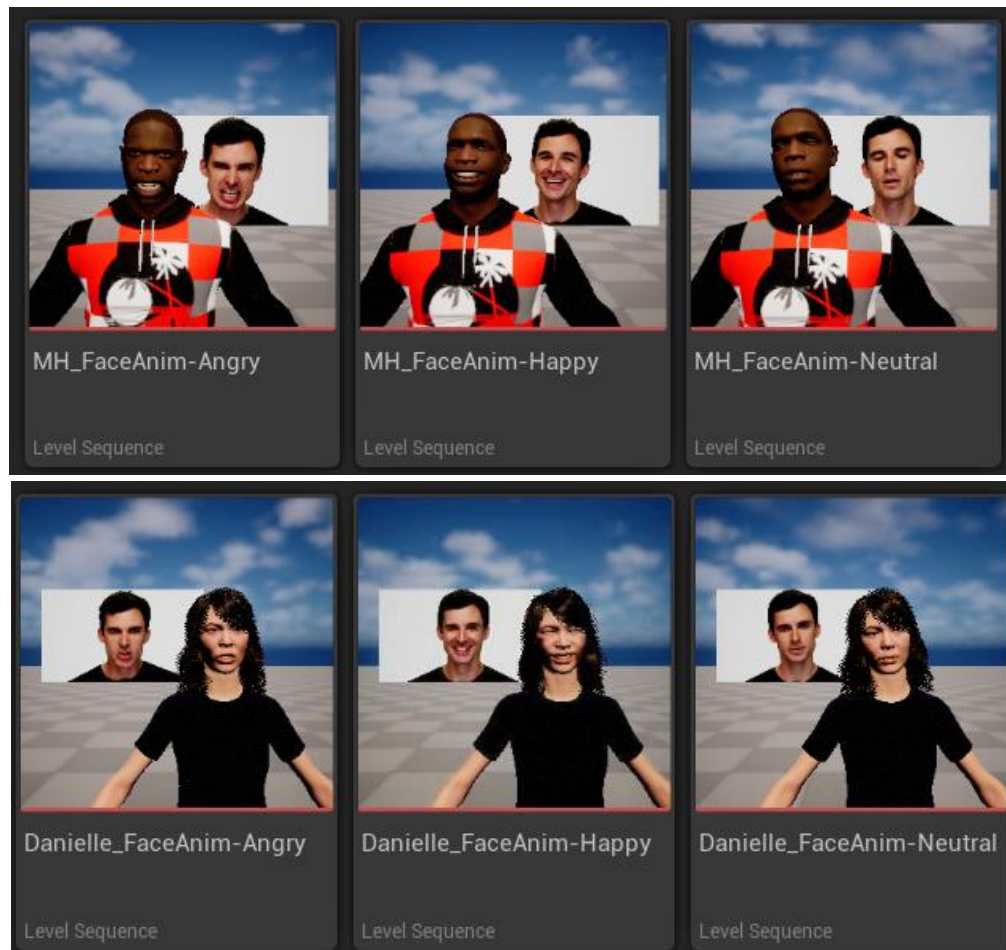
Demo 3 - Facial Animation

This demo focuses on generating and applying facial animation to MetaHuman characters using video-based performance capture. Facial animation was created using clips from the RAVDESS dataset, specifically from the *Video_Song_Actor_01* subset, which depicts a male actor singing short phrases with various emotional expressions. Three emotional states were selected for this demo: happy, neutral, and angry.

The selected video clips were imported into Unreal Engine and processed with the MetaHuman Animator system. For each clip, a MetaHuman Performance asset was created, and the input video was set to be a monocular facial capture source. The processing pipeline included facial tracking and solving, which resulted in animation data representing facial movements such as lip motion, eye movement, and brow deformation.



Once processed, the resulting animation was exported as an animation sequence and applied to two MetaHuman characters: a custom self-avatar and Danielle MetaHuman. Both characters were placed in a Level Sequence, which allowed for a direct visual comparison of the source video and the resulting facial animation. A plane displaying the original video was included alongside the characters to serve as a reference for determining the accuracy of the animation transfer.



Results

The MetaHuman Animator system successfully generated facial animation from the input video clips, capturing important aspects of the actor's performance such as mouth movement, eye behaviour, and overall emotional expression. On the custom MetaHuman, the transferred animation appeared powerful and expressive, with facial movements that closely matched the source performance. During the singing sequences, there was clear lip synchronisation, and emotional expressions such as happiness and anger were effectively conveyed.

When the same animation was applied to the Danielle MetaHuman, the overall motion was preserved, but the perceived intensity of the expressions was faintly reduced. While the timing and structure of the animation remained consistent, the emotional impact differed between the two characters. This contrast highlights the influence of facial proportions and rig behaviour on the final appearance of animation.



Evaluation and Reflection

The results show that MetaHuman Animator can produce high-quality facial animation from monocular video input, but it also has limitations in terms of consistency of emotion transfer across characters. One observation was the slight variation in expression intensity between the two MetaHumans. The custom MetaHuman showed strong and sometimes exaggerated expressions, whereas the Danielle MetaHuman showed more subdued facial movement. This disparity is most likely due to variations in facial structure, rig weighting, and how deformation is applied across different MetaHuman templates.

The nature of the source data also plays a role. The RAVDESS clips used in this demo show a male actor singing in various emotional states, which may not be directly applicable to characters with different facial characteristics. As a result, the same animation may appear overly expressive on one character and under expressive on another.

Despite these differences, the system maintained consistent temporal alignment and overall motion fidelity, implying that the underlying animation data was properly transferred. However, the variation in perceived emotion suggests that further tuning or character-specific adjustments may be required to achieve consistent results across multiple MetaHumans.

The demo demonstrates both the advantages and disadvantages of video-driven facial animation. While the technology allows for the rapid generation of detailed facial performances, the outcome is influenced by both the source material and the target character, so critical evaluation and refinement are essential components of the workflow.